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IT357 Quantum Computing Assignment 1

Code can be found in this Colab Notebook: [QuantumGates.ipynb](#)

Before - Grey

After - Red/Green

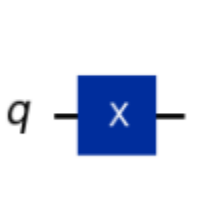
Quantum Gates

Pauli Gates: Pauli Gates are fundamental quantum logic gates that operate on qubits, the basic units of information in quantum computing. These gates are named after Wolfgang Pauli, a renowned physicist who made significant contributions to the field of quantum mechanics.

Pauli X

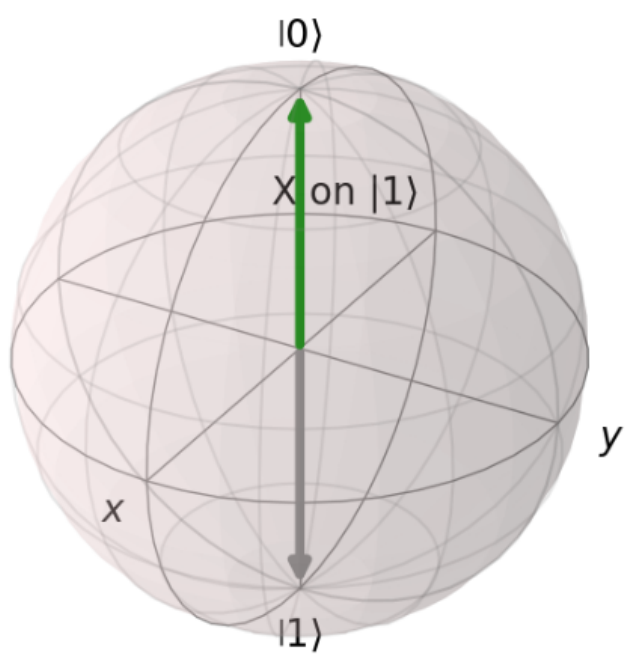
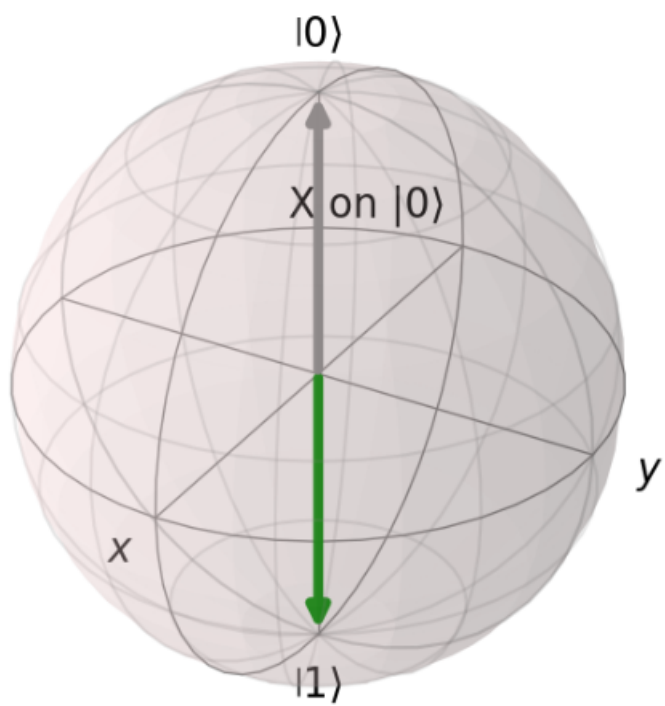
It flips a qubit's state:

- If the qubit is in state $|0\rangle$, it changes to $|1\rangle$
- If it's in state $|1\rangle$, it changes to $|0\rangle$.



Truth Table for X Gate:

	Input State	$ 0\rangle$	$ 1\rangle$
0	$ 0\rangle$	0.0+0.0j	1.0+0.0j
1	$ 1\rangle$	1.0+0.0j	0.0+0.0j



Pauli Y Gate

The Pauli-Y gate is a quantum gate that performs a π rotation around the Y-axis on the Bloch sphere.

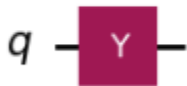
It flips the qubit similar to the X gate, but adds a phase of $\pm i$:

$$Y |0\rangle = i|1\rangle$$

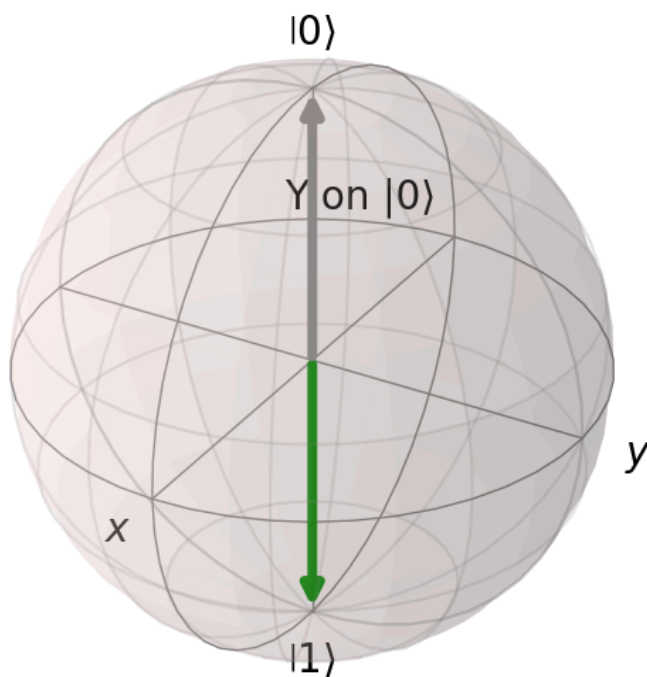
$$Y |1\rangle = -i|0\rangle$$

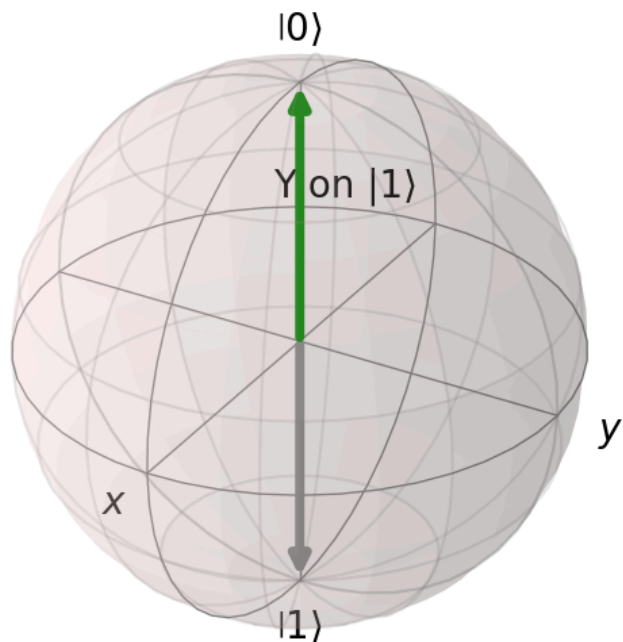
It's a combination of bit flip and phase shift.

180° rotation around the Y-axis of the Bloch sphere



	Input State	$ 0\rangle$	$ 1\rangle$	
0	$ 0\rangle$	$0.0+0.0j$	$0.0+1.0j$	
1	$ 1\rangle$	$0.0-1.0j$	$0.0+0.0j$	





Pauli Z Gate

It flips the phase of the $|1\rangle$ state.

$|0\rangle$ stays unchanged, while $|1\rangle$ becomes $-|1\rangle$.

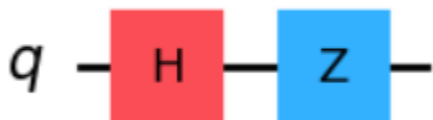
$|0\rangle$ stays at +Z pole

$|1\rangle$ stays at -Z pole (but rotates 180° in phase,

On the Bloch Sphere:

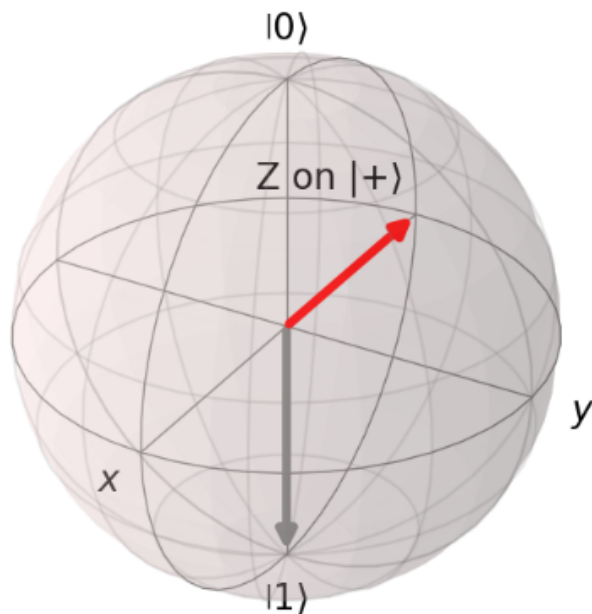
The Z gate rotates the qubit around the Z-axis by 180° .

It changes the relative angle of superpositions like $|+\rangle$ and $|-\rangle$.



Truth Table for Z gate:

Input	$ 0\rangle$	$ 1\rangle$
0	$ 0\rangle$ 1.0+0.0j	0.0+0.0j
1	$ 1\rangle$ 0.0+0.0j	-1.0+0.0j



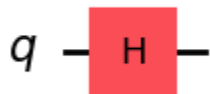
Hadamard Gate

The Hadamard gate (H gate) is a single-qubit quantum gate that creates superposition. If the input is 0, it turns it into both 0 and 1 at the same time.

If the input is 1, it also turns it into both 0 and 1, but with a minus sign on the 1 part.

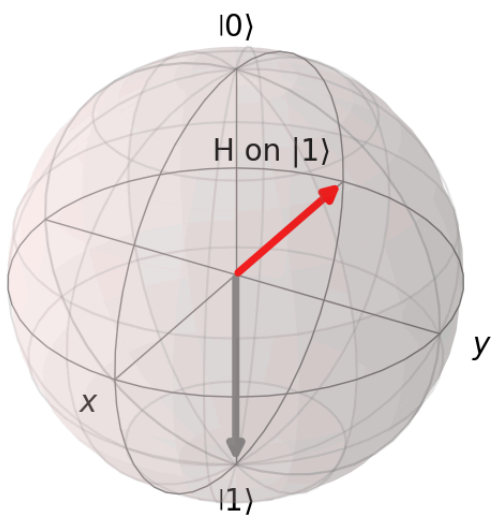
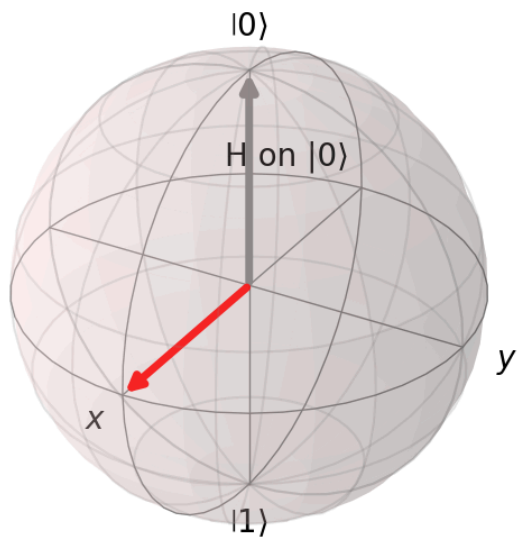
H applied to $|0\rangle$, the Bloch vector moves from North pole ($Z=1$) to the X-axis.

H applied to $|1\rangle$, it moves from South pole ($Z=-1$) to the $-X$ -axis.



Truth Table for H Gate:

	Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude	
0	$ 0\rangle$	$0.707107+0.000000j$	$0.707107+0.000000j$	
1	$ 1\rangle$	$0.707107+0.000000j$	$-0.707107+0.000000j$	



S Gate

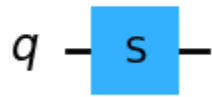
The S gate is a quantum phase gate that adds a 90° phase ($\pi/2$ radians) to the $|1\rangle$ state. It leaves $|0\rangle$ unchanged.

It multiplies $|1\rangle$ by i

On the Bloch Sphere:

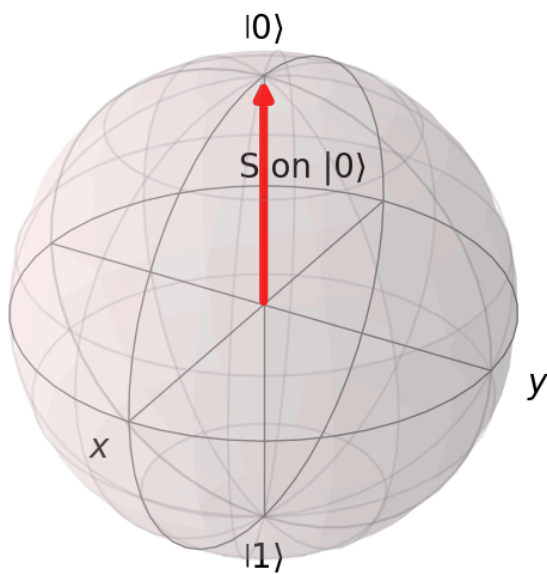
The magnitude of the state remains the same.

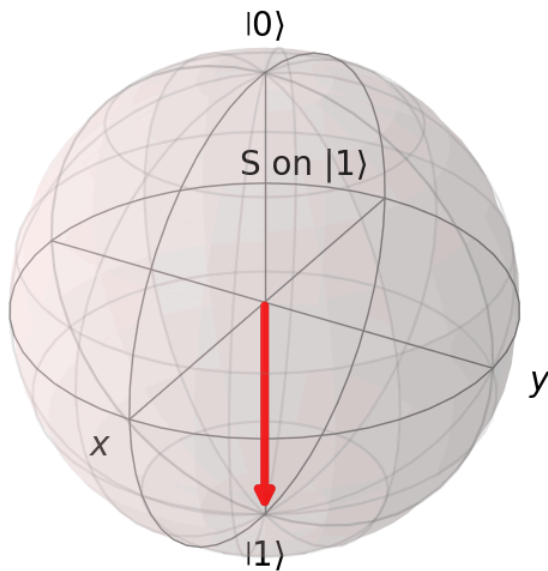
The phase of the $|1\rangle$ state is rotated around the Z-axis by 90° .



Truth Table for S Gate:

	Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude
0	$ 0\rangle$	$1.0+0.0j$	$0.0+0.0j$
1	$ 1\rangle$	$0.0+0.0j$	$0.0+1.0j$





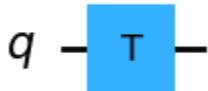
T Gate

The T gate is also known as the $\pi/8$ gate

It applies a phase shift of 45° ($\pi/4$ radians) to the qubit

It does not change the probability of measuring $|0\rangle$ or $|1\rangle$.

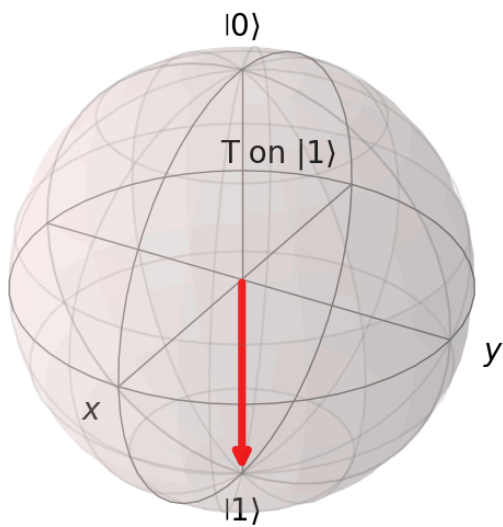
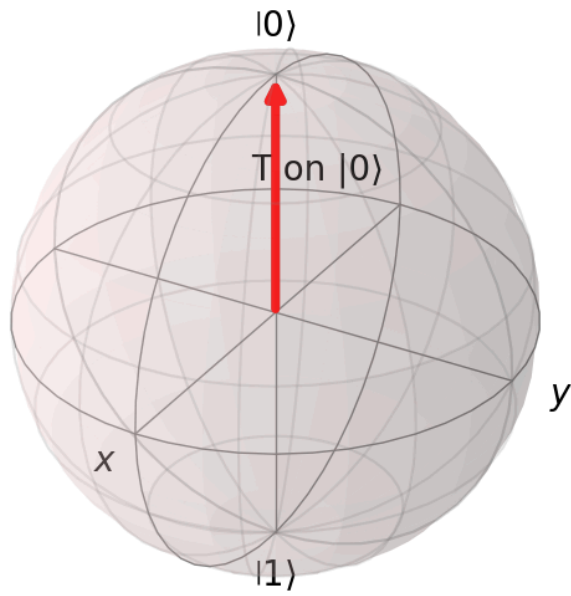
But it adds a phase to the $|1\rangle$ part of the qubit:



Truth Table for T Gate:

	Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude
0	$ 0\rangle$	$1.0+0.0j$	$0.000000+0.000000j$
1	$ 1\rangle$	$0.0+0.0j$	$0.707107+0.707107j$





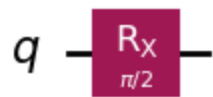
RX Gate

The $RX(\theta)$ gate rotates the qubit around the X-axis of the Bloch sphere by an angle θ (in radians).

$RX(\pi/2)$:

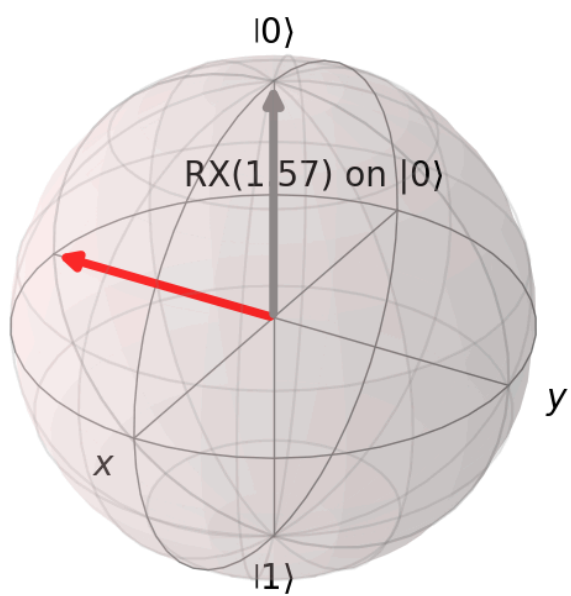
It changes the qubit state by rotating it 90° around the X-axis.

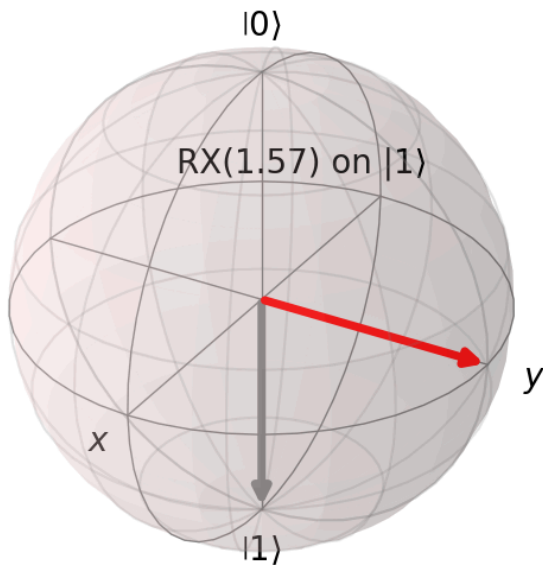
This means if the qubit starts at $|0\rangle$ after $RX(\pi/2)$, it will move towards the equator



Truth Table for $R_X(1.57)$ Gate:

	Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude
0	$ 0\rangle$	$0.707107+0.000000j$	$0.000000-0.707107j$
1	$ 1\rangle$	$0.000000-0.707107j$	$0.707107+0.000000j$





R_Y Gate

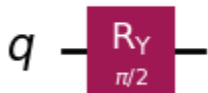
The R_y gate is a rotation gate that rotates a qubit around the Y-axis of the Bloch sphere.

After applying $R_y(\pi/2)$ on input $|0\rangle$

Qubit rotates downward by $\pi/3$ radians (90°) along the Y-axis

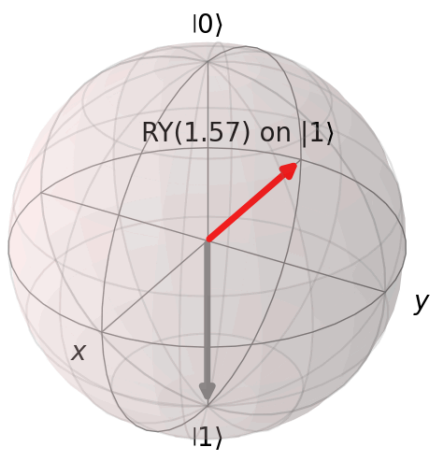
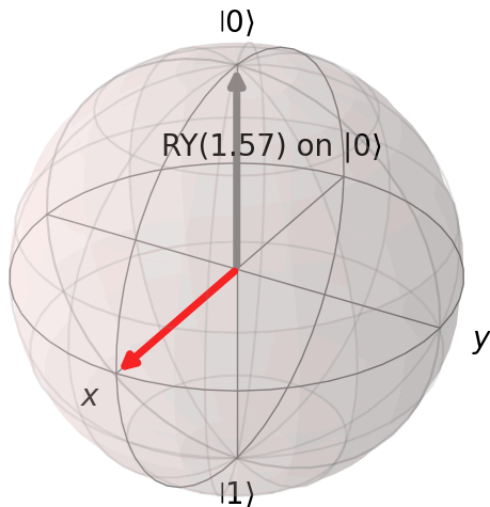
After applying $R_y(\pi/2)$ on input $|1\rangle$

Qubit rotates upward by $\pi/3$ radians (90°) along the Y-axis



Truth Table for $R_Y(1.57)$ Gate:

	Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude	
0	$ 0\rangle$	$0.707107+0.000000j$	$0.707107+0.000000j$	
1	$ 1\rangle$	$-0.707107+0.000000j$	$0.707107+0.000000j$	



RZ Gate

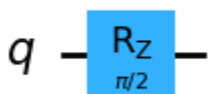
$RZ(\theta)$ gate rotates a qubit around the Z-axis of the Bloch sphere.

It doesn't change the $|0\rangle$ or $|1\rangle$ probabilities, only the relative phase.

$RZ(\pi/2)$ applies a 90° rotation around Z-axis.

$|0\rangle$ lies at the north pole of the Bloch sphere. RZ only affects phase, so nothing changes visually

$|1\rangle$ lies at the south pole. Again, only the phase changes, so no Bloch vector movement is visible

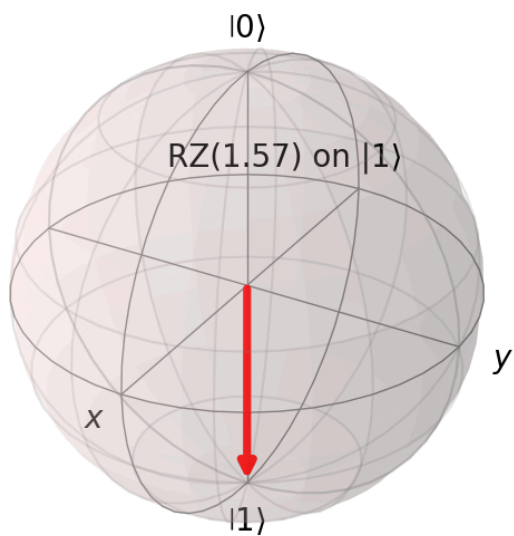
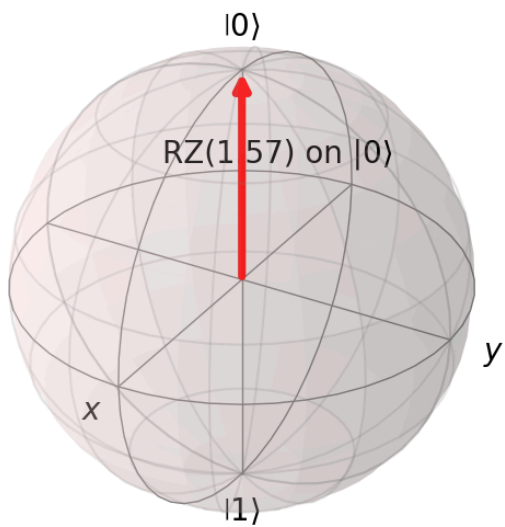


Truth Table for RZ(1.57) Gate:

Input State	$ 0\rangle$ amplitude	$ 1\rangle$ amplitude
0	$ 0\rangle \quad 0.707107-0.707107j$	$0.000000+0.000000j$
1	$ 1\rangle \quad 0.000000+0.000000j$	$0.707107+0.707107j$



Before after overlap



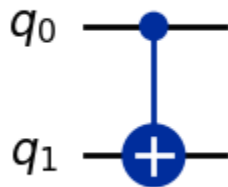
CNOT gate

The CNOT (Controlled-NOT) gate is a two-qubit gate.

If the control qubit is $|1\rangle$, it flips the target qubit.

If the control is $|0\rangle$, the target remains unchanged.

CNOT can't be shown on a Bloch sphere because it's a 2-qubit gate, and the Bloch sphere represents only single-qubit states.

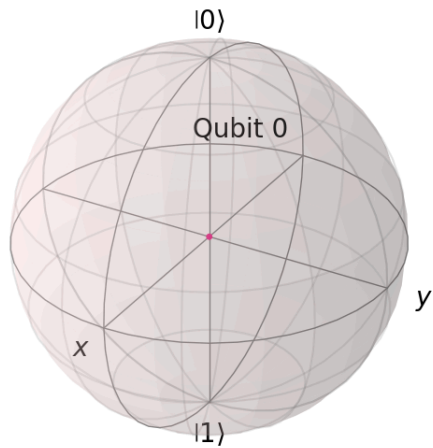


Truth Table for CNOT Gate:

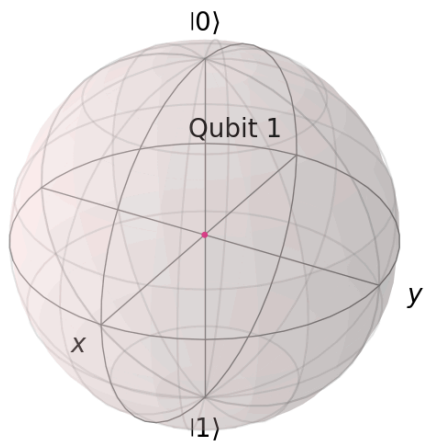
	Input State	$ 00\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$
0	$ 00\rangle$	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
1	$ 01\rangle$	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j
2	$ 10\rangle$	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j
3	$ 11\rangle$	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j

For the Bloch sphere, since CNOT entangles qubits, single-qubit Bloch spheres only show reduced info for one qubit.

Bloch sphere of qubit 0 after CNOT on $|+0\rangle$:

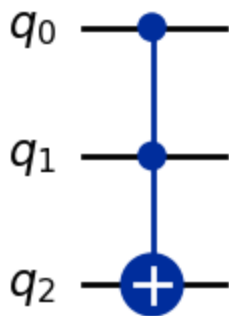


Bloch sphere of qubit 1 after CNOT on $|+0\rangle$:



Controlled CNOT / Toffoli Gate

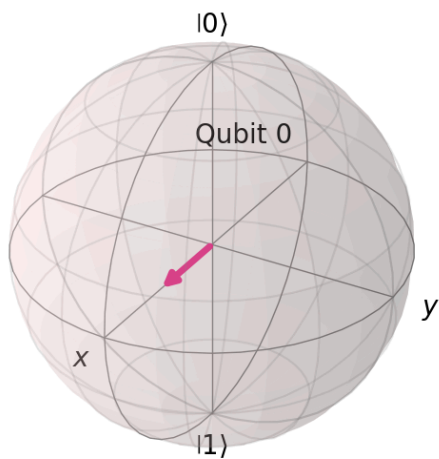
The Toffoli gate (Controlled-CNOT or CCX) flips the target qubit if both control qubits are 1. The Bloch sphere represents single-qubit states only. The CCX gate involves three qubits, so it can't be visualized on a single Bloch sphere.



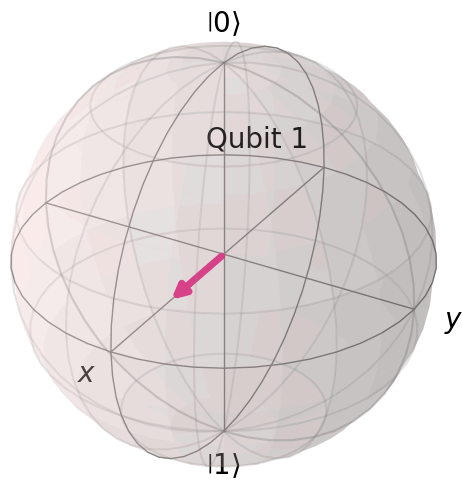
Truth Table for CCNOT (Toffoli) Gate:

	Input State	000⟩	001⟩	010⟩	011⟩	100⟩	101⟩	110⟩	111⟩
0	000⟩	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
1	001⟩	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
2	010⟩	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
3	011⟩	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j
4	100⟩	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
5	101⟩	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j
6	110⟩	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j
7	111⟩	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j

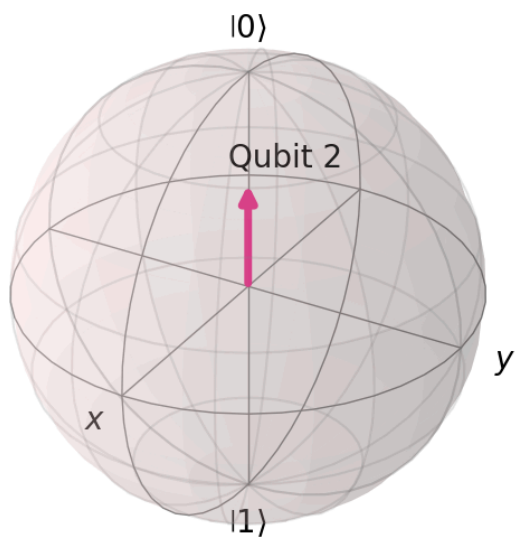
Bloch sphere of qubit 0 after CCNOT on $|++0\rangle$:



Bloch sphere of qubit 1 after CCNOT on $|++0\rangle$:



Bloch sphere of qubit 2 after CCNOT on $|++0\rangle$:



SWAP gate:

the SWAP gate is a 2-qubit operation, and the Bloch sphere only represents 1-qubit states.

Hence, it cannot visualize multi-qubit entanglement or state exchange

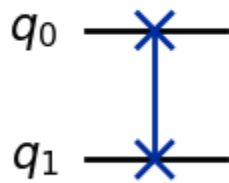
The SWAP gate exchanges the states of two qubits. It's a two-qubit gate, made of three CNOT gates.

$$|00\rangle \rightarrow |00\rangle$$

$$|01\rangle \rightarrow |10\rangle$$

$$|10\rangle \rightarrow |01\rangle$$

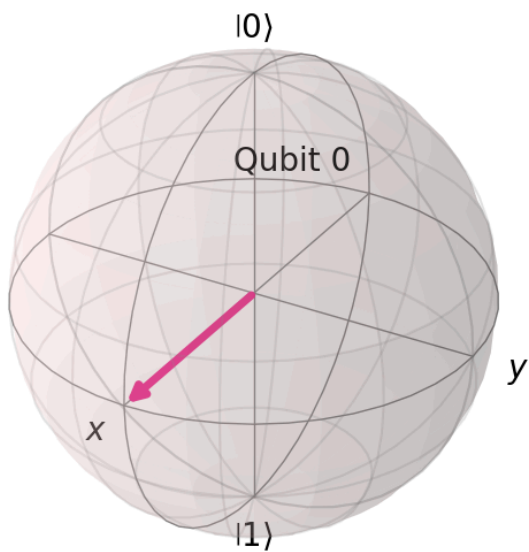
$$|11\rangle \rightarrow |11\rangle$$



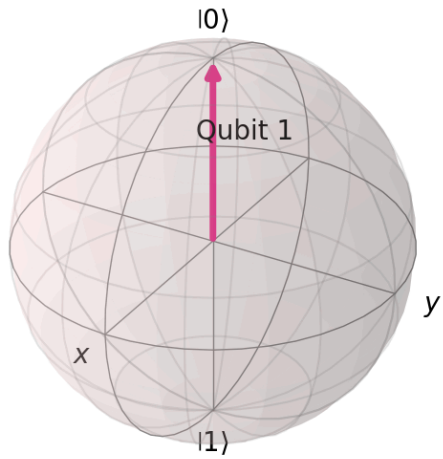
Truth Table for SWAP Gate:

	Input State	$ 00\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$
0	$ 00\rangle$	1.0+0.0j	0.0+0.0j	0.0+0.0j	0.0+0.0j
1	$ 01\rangle$	0.0+0.0j	1.0+0.0j	0.0+0.0j	0.0+0.0j
2	$ 10\rangle$	0.0+0.0j	0.0+0.0j	1.0+0.0j	0.0+0.0j
3	$ 11\rangle$	0.0+0.0j	0.0+0.0j	0.0+0.0j	1.0+0.0j

Bloch sphere of qubit 0 after SWAP on $|+0\rangle$:



Bloch sphere of qubit 1 after SWAP on $|+0\rangle$:

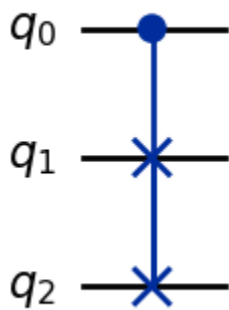


Controlled SWAP gate (Fredkin gate)

Swaps qubit 1 and 2 only if qubit 0 = $|1\rangle$ (control qubit).

Otherwise, does nothing.

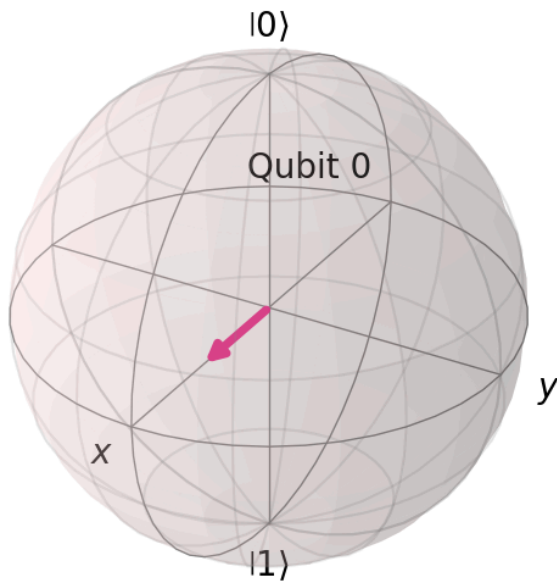
Controlled gates can't be represented on the Bloch sphere because they act on multiple qubits and can create entanglement



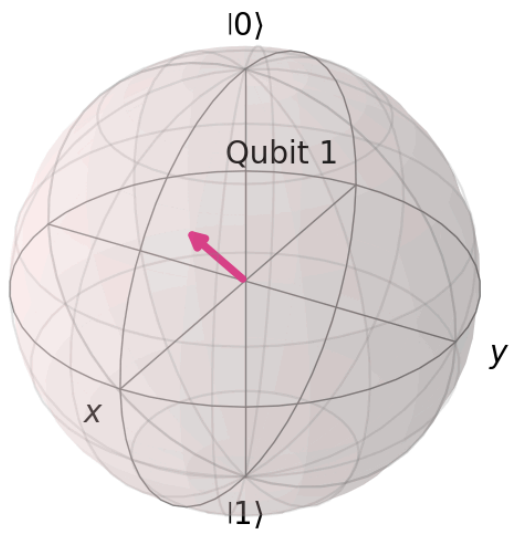
Truth Table for Controlled SWAP (Fredkin) Gate:

	Input State	$ 000\rangle$	$ 001\rangle$	$ 010\rangle$	$ 011\rangle$	$ 100\rangle$	$ 101\rangle$	$ 110\rangle$	$ 111\rangle$
0	$ 000\rangle$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
1	$ 001\rangle$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
2	$ 010\rangle$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
3	$ 011\rangle$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$
4	$ 100\rangle$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
5	$ 101\rangle$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
6	$ 110\rangle$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$
7	$ 111\rangle$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$0.0+0.0j$	$1.0+0.0j$

Bloch sphere of qubit 0 after Controlled SWAP on $|++0\rangle$:



Bloch sphere of qubit 1 after Controlled SWAP on $|++0\rangle$:



Bloch sphere of qubit 2 after Controlled SWAP on $|++0\rangle$:

